

Lesson 1-1

Development of Information Technology and Social Transformation

Learning Objectives

- Explain the major stages in the development of information technology and their impact on society.
- Give examples of social changes and emerging technologies brought about by information technology, and explain their characteristics.

On an ordinary day, a student in Egypt checks messages on an SNS app, pays for breakfast with a cashless app, joins a lesson through online learning, and orders a book from an e-commerce (EC) shop. Twenty years ago, most of this was not possible. Information technology has developed through a series of stages — from the first computers to cloud computing — and at each stage it has changed the way people communicate, work, learn, and pay.

? Guiding question: How has information technology developed through its major stages, and how has each stage changed society?

✎ Explore (in pairs)

With a partner, look at the five time periods in the table below before you read on. For each period, name one device or service from that era that you still use or hear about today. Then predict which stage changed daily life the most, and give one reason.

Key Fact:
At each stage, information technology introduced a new technology or service and also changed how society communicates, works, and does business.

Key Concepts:

- Moore's Law
- SNS
- e-commerce
- remote work
- online learning
- cashless payment
- edge computing
- autonomous driving
- AR / VR
- quantum computing

📌 Point!

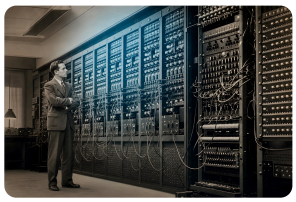
1. The History of Information Technology (IT)

(1) The major stages in the development of information technology are summarized in the table below.

Time period	Major technologies and events	Impact on society
1940s-60s	Birth of the computer (ENIAC, vacuum tubes)	Mainly used for military and scientific computation
1970s-80s	Spread of personal computers (PCs)	Beginning of personal computer use
1990s	Commercialization of the Internet; the Web	Globalization of information; spread of email
2000s	Rise of smartphones (iPhone, etc.)	Explosive spread of mobile Internet
2010s onward	Spread of cloud computing	Large-scale data analysis and AI; "IT as a service" becomes widespread



IT advanced in stages, each smaller and more connected.

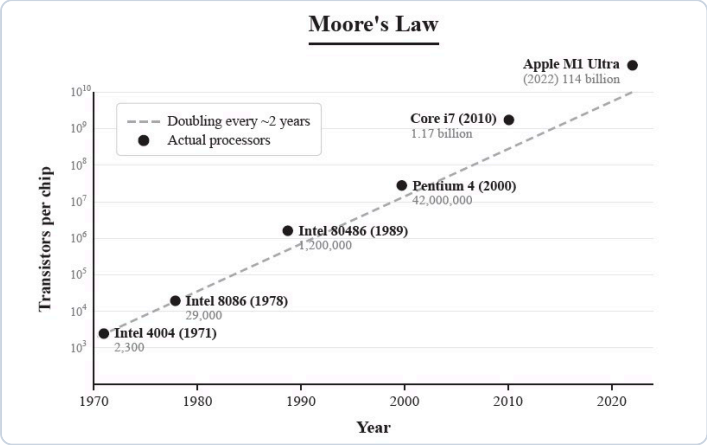


The first computers filled a whole room.

(2) **(Moore's Law)**...the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years."

This remained largely accurate for many years, and computing power increased dramatically.

However, in recent years, miniaturization of transistors has been approaching a physical limit. If circuits are made any smaller, problems arise such as the quantum tunneling effect, in which electrons slip through barriers, and leakage current, in which current escapes unintentionally, making it difficult to achieve both higher performance and lower power consumption at the same time. In response, new directions for performance improvement are being pursued, such as parallel processing using multiple processor cores, and quantum computers based on the principles of quantum mechanics.



Moore's Law: transistor count over time

2. Social Changes Resulting from Information Technology

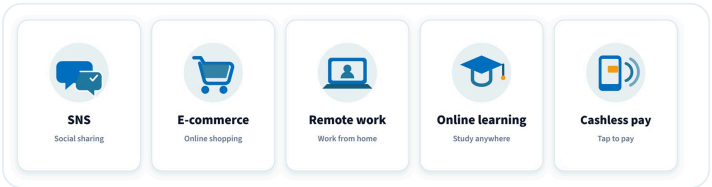
(1) **(SNS (Social Networking Service))**...services that allow users to connect with each other and post and share information. They are highly effective at spreading information rapidly.

(2) **(E-commerce (EC))**...buying and selling goods and services through the Internet.
e.g., online shops such as Amazon and eBay

(3) **(Remote work)**...a working style in which work is performed from home or other remote locations using the Internet.

(4) **(Online learning)**...a learning style in which classes and study materials are delivered using the Internet.

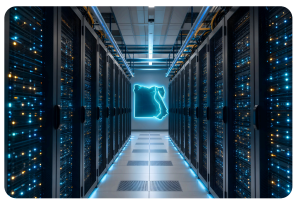
(5) **(Cashless payment)**...a system for making payments using electronic money, QR codes, etc., without using cash.
e.g., credit cards, debit cards, mobile payment apps



Five social changes driven by information technology.

KEY TERMS

Moore's Law — the observation that the number of transistors on a chip roughly doubles about every two years.



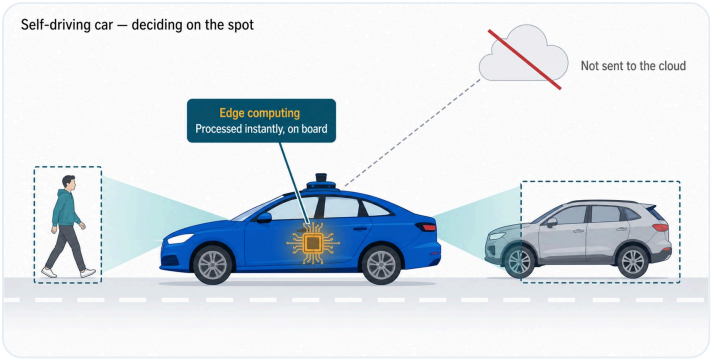
This is "the cloud" — real machines in a real place.

PAUSE & THINK

Of these five changes, which would be hardest to give up — and why?

3. Notable Emerging Technologies

(1) **(Autonomous driving)**...a technology that uses AI to drive a vehicle without human operation. It uses cameras and sensors to recognize the surroundings, makes driving decisions, and controls a vehicle. Because a delay of even 0.1 seconds can lead to an accident, edge computing, in which processing is performed instantly on the vehicle itself rather than sending data to the cloud for judgment, is used.

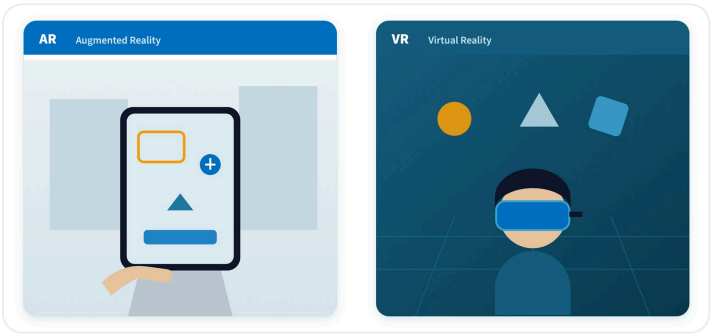


A self-driving car decides on board (edge computing), without waiting for the cloud.

(2) AR / VR

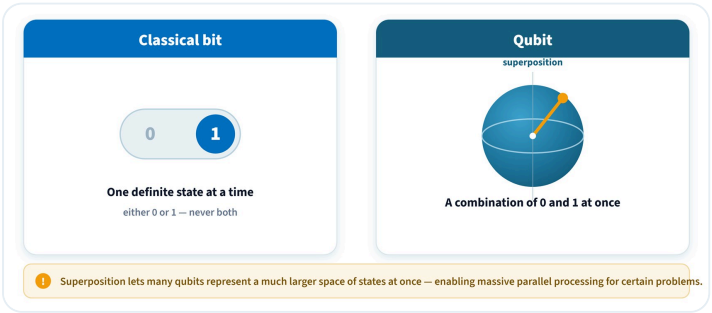
① **(AR (Augmented Reality))**...a technology that overlays digital information on real-world images.

② **(VR (Virtual Reality))**...a technology that allows users to immerse themselves in a virtual space generated by a computer.



AR adds digital layers to the real world; VR replaces it with a virtual one.

(3) **(Quantum computing)**...a technology expected to dramatically speed up computations that are difficult or impossible for traditional computers, by using the principles of quantum mechanics.



A classical bit holds one state; a qubit uses superposition.

KEY TERMS

edge computing — processing data on the device itself, instantly, instead of sending it to the cloud.

PAUSE & THINK

In autonomous driving, why is it necessary to process data instantly on the vehicle side using edge computing, rather than sending the data to the cloud for judgment? Explain your answer.

✎ Worked Example

(1) From the following options (A–D), choose the one that lists the stages of information technology (IT) development in the correct chronological order.

A Birth of the computer → Rise of smartphones → Commercialization of the Internet → Spread of cloud computing

B Birth of the computer → Commercialization of the Internet → Rise of smartphones → Spread of cloud computing

C Commercialization of the Internet → Birth of the computer → Spread of cloud computing → Rise of smartphones

D Rise of smartphones → Commercialization of the Internet → Birth of the computer → Spread of cloud computing

(2) Mark each statement (A–D) with ○ if true or × if false.

A Moore's Law is the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years."

B Moore's Law has been said to be approaching a physical limit in recent years.

C SNS is highly effective at spreading information rapidly.

D E-commerce (EC) refers to purchasing goods at physical stores using cash.

(3) Match each description (a–c) with the most appropriate technology from the choices below (A–C).

a A technology that overlays digital information on real-world images

b A technology that uses AI to drive a vehicle without human operation

c A technology that allows users to immerse themselves in a virtual space generated by a computer

[Choices] **A** Autonomous driving **B** AR **C** VR

✓ Solution

(1) The correct order is "Birth of the computer (1940s-60s) → Commercialization of the Internet (1990s) → Rise of smartphones (2000s) → Spread of cloud computing (2010s onward)." Therefore, **B**.

(2)

A Correct definition of Moore's Law. Therefore, ○.

B It has been said to be approaching a physical limit in recent years. Therefore, ○.

C SNS is a service highly effective at spreading information rapidly. Therefore, ○.

D E-commerce refers to buying and selling through the Internet, not purchases at physical stores using cash. Therefore, ×.

KEY TERMS

Moore's Law — transistors per chip roughly double every two years.

edge computing — data processed on the device itself, instantly.

(3)

a: Overlaying digital information on real-world images is AR (Augmented Reality). **B**

b: Driving a vehicle without human operation is autonomous driving. **A**

c: Immersing oneself in a virtual space is VR (Virtual Reality). **C**

Try

1 Answer the following questions.

1. What is the name of the empirical observation that "the number of transistors on an integrated circuit doubles approximately every two years"?
2. What is the term for the buying and selling of goods and services using the Internet?
3. What is the term for the working style in which one works from home or other remote locations using the Internet?
4. What is the term for the system for making payments using electronic money, QR codes, etc., without using cash?
5. What is the term for the technology that uses AI to drive a vehicle without human operation?

2 Answer the following questions.

(1) For each of a–d, indicate which of the following is most closely related: **A** Changes in daily life, **B** Changes in industry and the economy, **C** Changes in healthcare and education.

a Taking classes through online learning.

b Paying for purchases using a mobile payment app on a smartphone.

c Sharing photos with friends on SNS.

d A company introducing a work-from-home system.

(2) From the following options (A–D), choose the one that is NOT an appropriate description of an emerging technology.

A Autonomous driving uses AI to drive a vehicle without human operation.

B AR is a technology that overlays digital information on real-world images.

C VR is a technology that dramatically improves the processing speed of a computer.

D Quantum computing is expected to speed up computations that are difficult for traditional computers.

EXAM-STYLE QUESTION

Analyze how the spread of cloud computing (from the 2010s onward) has changed the way information technology is used. **[6] In your answer, refer to:** large-scale data analysis, AI, and "IT as a service".

KEY TERMS

remote work — working from home or another remote location using the Internet.

online learning — classes and materials delivered over the Internet.

⚙ Think as an Engineer — Investigate & Decide

Start from the advantage and concern you named in the Pause & Think note, then investigate further and decide.

1 · Collect data. Survey ten classmates: do they usually pay with cash or with a cashless method (card, mobile app, or QR code)? Record the results in a table and identify the most common response.

2 · Analyze the stakeholders. Using your survey and your own reasoning, note one benefit and one drawback of a fully cashless society for each group below.

Group	Benefit	Drawback
A customer paying		
A small shop owner		
A person with no bank card or smartphone		

3 · Decide. Using your survey and your table, decide: should your community move toward cashless payment? Recommend one step and give two reasons.

Stuck? Start with the customer — one benefit is speed of payment; one drawback is needing a card or a phone.

💡 PAUSE & THINK

Cashless payment is spreading in many countries. If a fully cashless society were realized, choose one advantage and one possible concern, and briefly explain the reason for each.

🌍 In a New Context

A village that has never had Internet access is connected to high-speed Internet and cashless payment for the first time. Using the stages and social changes you studied, predict two ways daily life in the village will change, and identify one new problem the village may face.

KEY TERMS

cloud computing — IT delivered as a service over the Internet.
e-commerce — buying and selling through the Internet.

❓ Lesson Question — Answered

The lesson question asked how information technology developed and how each stage changed society. Information technology advanced through a series of stages rather than in a single step, and each stage changed much more than the technology itself. As computing spread, information became global and then mobile, many everyday activities moved online, and information technology grew into a service that supports large-scale data analysis and AI. Each stage also brought social changes in how people communicate, work, learn, and make payments. Overall, the development of information technology is not only about faster machines; at every stage it has reshaped daily life, industry, and the way society handles information.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Computers were invented in the 1940s and were initially used mainly for military and scientific computation. Later, in the 1970s and 1980s, (a) became widespread, allowing individuals to use computers. In the 1990s, (b) was commercialized, advancing the globalization of information. In the 2000s, (c) emerged, leading to the explosive spread of mobile Internet.

1. Fill in the blanks (a)–(c).
2. What technology, which spread from the 2010s onward, supports large-scale data analysis and the use of AI?
3. According to Moore's Law, about how often does the number of transistors on an integrated circuit double?

3 Answer the following questions.

(1) From the following options (A–F), choose all that are emerging technologies.

A Email **B** Autonomous driving **C** Personal computers **D** AR **E** VR **F** Quantum computing

(2) From the following options (A–D), choose the one that most appropriately describes the characteristics of SNS.

A A service that allows users to connect with each other and post and share information, and is highly effective at spreading information rapidly.

B A service for buying and selling goods and services through the Internet.

C A working style in which one works from home or other remote locations.

D A system for making payments using electronic money or QR codes without using cash.

REFLECT & CHALLENGE

Reflect: Which stage of information technology do you think will matter most in the next ten years? Give one reason. Was your prediction at the start of the lesson correct? What changed your mind? **Challenge:** Choose one emerging technology from this lesson (autonomous driving, AR, VR, or quantum computing). Suggest one way it could help solve a real problem, and state one risk.

KEY TERMS

SNS — connects users to post and share information.
cashless payment — payment without cash (e-money, QR).

Key takeaway

Remember:
Information technology developed in stages — computers, the Internet, smartphones, and cloud computing. At each stage it introduced a new technology or service and also changed how society communicates, works, learns, and pays.



Learning Objectives

- Explain what AI is.
- Explain how generative AI is positioned within AI technologies.

When you use a phone, AI is often at work. A spam filter sorts your email, a store recommends products you might buy, a translation app changes one language into another, and ChatGPT generates text from a prompt. These are all examples of AI, but they are not all the same kind. AI includes machine learning, within machine learning is deep learning, and today's generative AI systems are built on deep learning — a series of nested technologies, each more specialized than the last.

? Guiding question: What is AI, and how are machine learning, deep learning, and generative AI related?

✎ Explore · In Pairs

With a partner, list three tasks a phone or computer does for you that seem to need "intelligence" (for example, sorting spam, recommending a video, or translating text). Then predict: does the computer follow fixed rules, or does it learn from examples? Give one reason.

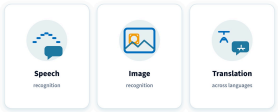
📌 Point!

1. What is Artificial Intelligence (AI)?

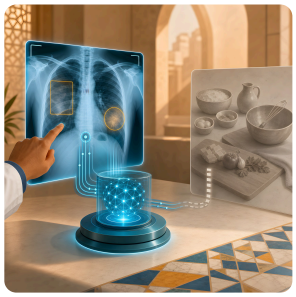
(1) **(AI (Artificial Intelligence))**…a general term for technologies that reproduce or perform intelligent human behavior (learning, reasoning, judgment, etc.) on a computer.
e.g., speech recognition, image recognition, translation

2. The Relationship between AI and Machine Learning

(1) Within AI technologies, there is machine learning; within machine learning, there is deep learning; and within deep learning, there is generative AI. The broad field of AI includes increasingly specialized technologies as a series of nested categories.



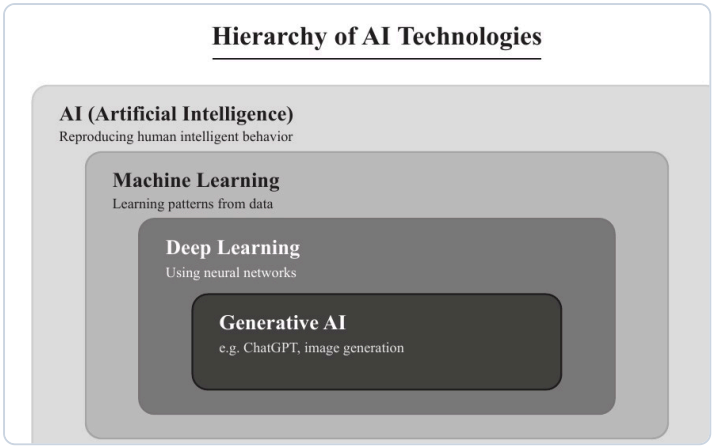
Three everyday AI examples: speech, image, translation.



Today's AI is narrow — expert at one task only.

💡 THINK IT THROUGH

Spam filters and product recommendations are both examples of machine learning, but the tasks they perform are completely different. From the perspective of how AI works, explain what these two applications have in common.

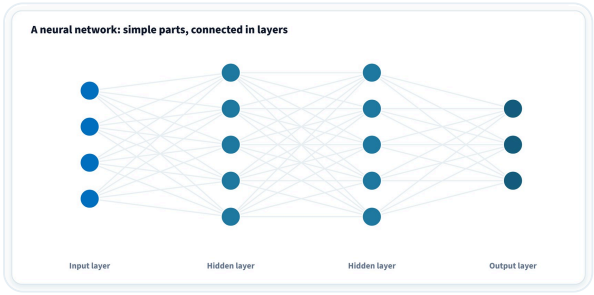


Hierarchy of AI Technologies

(2) **(Machine Learning)**…one of the learning technologies that makes AI work. It learns patterns from data to make predictions and judgments.
e.g., spam filters, product recommendations

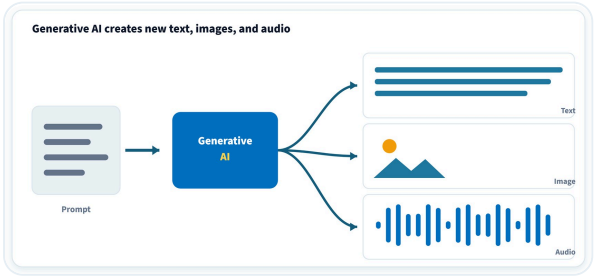
(3) **(Deep Learning)**…an advanced technology within machine learning that uses neural networks. It learns complex patterns using large-scale data.
e.g., image analysis for autonomous driving, speech synthesis

(4) **(Neural network)**…a system modeled after the workings of the nerve cells (neurons) of the human brain. By connecting many components, it learns from data and becomes capable of making complex judgments. It is the core technology supporting recent advances in AI.

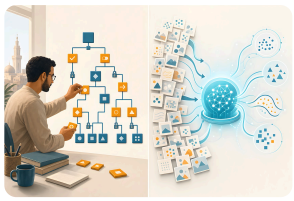


A neural network connects many simple components.

(5) **(Generative AI)**…AI technology that uses deep learning to generate new data (text, images, audio, programs, etc.).
e.g., ChatGPT, image generation AIs



Generative AI creates new text, images, and audio.



Fixed hand-written rules versus rules learned from data.

PAUSE & THINK

Deep learning needs large-scale data to learn. Why might an AI struggle with something it has rarely seen in its data?

KEY TERMS

neural network — a system modeled after the nerve cells (neurons) of the human brain, used to learn from data and make complex judgments.

KEY TERMS

generative AI — AI that uses deep learning to generate new data such as text, images, and audio.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Machine learning is one of the technologies that makes AI work.

B Deep learning is a completely different technology from machine learning.

C Generative AI is a technology that uses deep learning to generate new data.

D AI and machine learning have the same meaning.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A technology that learns patterns from data to make predictions and judgments

b A technology that uses neural networks to learn complex patterns

c AI technology that generates new data such as text and images

[Choices] **A** Machine learning **B** Deep learning **C** Generative AI

✓ Solution

(1)

A Machine learning is one of the learning technologies that makes AI work. Therefore, ○.

B Deep learning is a type of machine learning (an advanced technology within machine learning). Therefore, ×.

C Correct description of generative AI. Therefore, ○.

D AI is a broad concept, and machine learning is one of the technologies within it. They do not have the same meaning. Therefore, ×.

(2)

a: Learning patterns from data is machine learning. **A**

b: Using neural networks is deep learning. **B**

c: Generating new data is generative AI. **C**

KEY TERMS

machine learning — learns patterns from data.

deep learning — uses neural networks on large-scale data.

Try

1 Answer the following questions.

1. What is the general term, expressed by a two-letter abbreviation, for technologies that reproduce or perform intelligent human behavior on a computer?
2. What is the term for the learning technology that makes AI work by learning patterns from data to make predictions and judgments?
3. What is the term for the advanced technology within machine learning that uses neural networks?
4. What is the term for the AI technology that uses deep learning to generate new data?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that correctly describes the relationship between AI, machine learning, deep learning, and generative AI.

A AI > Deep Learning > Machine Learning > Generative AI

B Machine Learning > AI > Generative AI > Deep Learning

C AI > Machine Learning > Deep Learning > Generative AI

D Generative AI > Deep Learning > Machine Learning > AI

(2) From the following options (A–D), choose the one that is NOT an appropriate example of generative AI.

A ChatGPT

B Image generation AI

C Spam filter

D Audio generation AI

Think as an Engineer — Investigate & Decide

Starting from the Think It Through note about hallucination, investigate further and decide.

1 · Investigate. Find one documented example where a generative AI gave an answer that sounded correct but was actually wrong (a "hallucination"). Record its source — a news article, an official report, or a prompt you tested yourself with a screenshot — and note where you found it. Do not rely on an example recalled from memory without a source.

2 · Weigh it up. Note one benefit and one risk of using generative AI to write a school report — for the student, and for the teacher.

3 · Decide. Should generative AI be allowed for homework in your school? Recommend one rule and give two reasons.

EXAM-STYLE QUESTION

Analyze the relationship between AI, machine learning, deep learning, and generative AI. **[6] In your answer:** use the idea of nested categories, and give one example of each.

THINK IT THROUGH

Generative AI can produce text that sounds plausible but is factually incorrect (a "hallucination"). Explain why it is dangerous to use such output as the answer to a school report as-is, and state what learners should do instead.

Stuck? One benefit is speed; one risk is that the answer may be factually incorrect, so it must be checked.

🌍 In a New Context

A farmer wants to use AI to distinguish healthy crops from diseased ones in photographs. Using what you learned about deep learning and image analysis, explain what the AI would need to learn from, and identify one reason its judgment might be wrong.

❓ Lesson Question — Answered

The lesson question asked what AI is and how machine learning, deep learning, and generative AI are related. AI is a general term for technologies that reproduce or perform intelligent human behavior on a computer, and machine learning, deep learning, and generative AI are increasingly specialized fields nested within it rather than parallel to it. The key idea from this lesson is that these are not four separate technologies to learn in isolation, but one broad field understood at different levels of specialization.

KEY TERMS

deep learning — uses neural networks to learn complex patterns.

generative AI — uses deep learning to generate new data.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

The general term for technologies that reproduce or perform intelligent human behavior on a computer is (a). One of the learning technologies that makes (a) work by learning patterns from data is (b). Within (b), an advanced technology that uses neural networks is (c). Furthermore, the technology that uses (c) to generate new data is (d).

1. Fill in the blanks (a)–(d).

⭐ REFLECT & CHALLENGE

Reflect: Before this lesson, did you think AI was one single technology? How has your idea changed? Which of the four — AI, machine learning, deep learning, generative AI — was hardest to tell apart, and why?

Challenge: Generative AI can create text, images, and audio. Suggest one helpful use and one possible misuse, and say how the misuse could be reduced.

3 Answer the following questions.

(1) For each example (a–d), indicate which of the following is most closely related: **A** Machine learning, **B** Deep learning, **C** Generative AI.

a Automatically sorting spam emails

b Recognizing pedestrians on the road in autonomous driving

c Automatically generating new images from text

d Recommending products based on a user's purchase history

(2) From the following options (A–D), choose the one that most appropriately describes the relationship between AI, machine learning, deep learning, and generative AI.

A Machine learning is a type of AI, and deep learning is a type of machine learning.

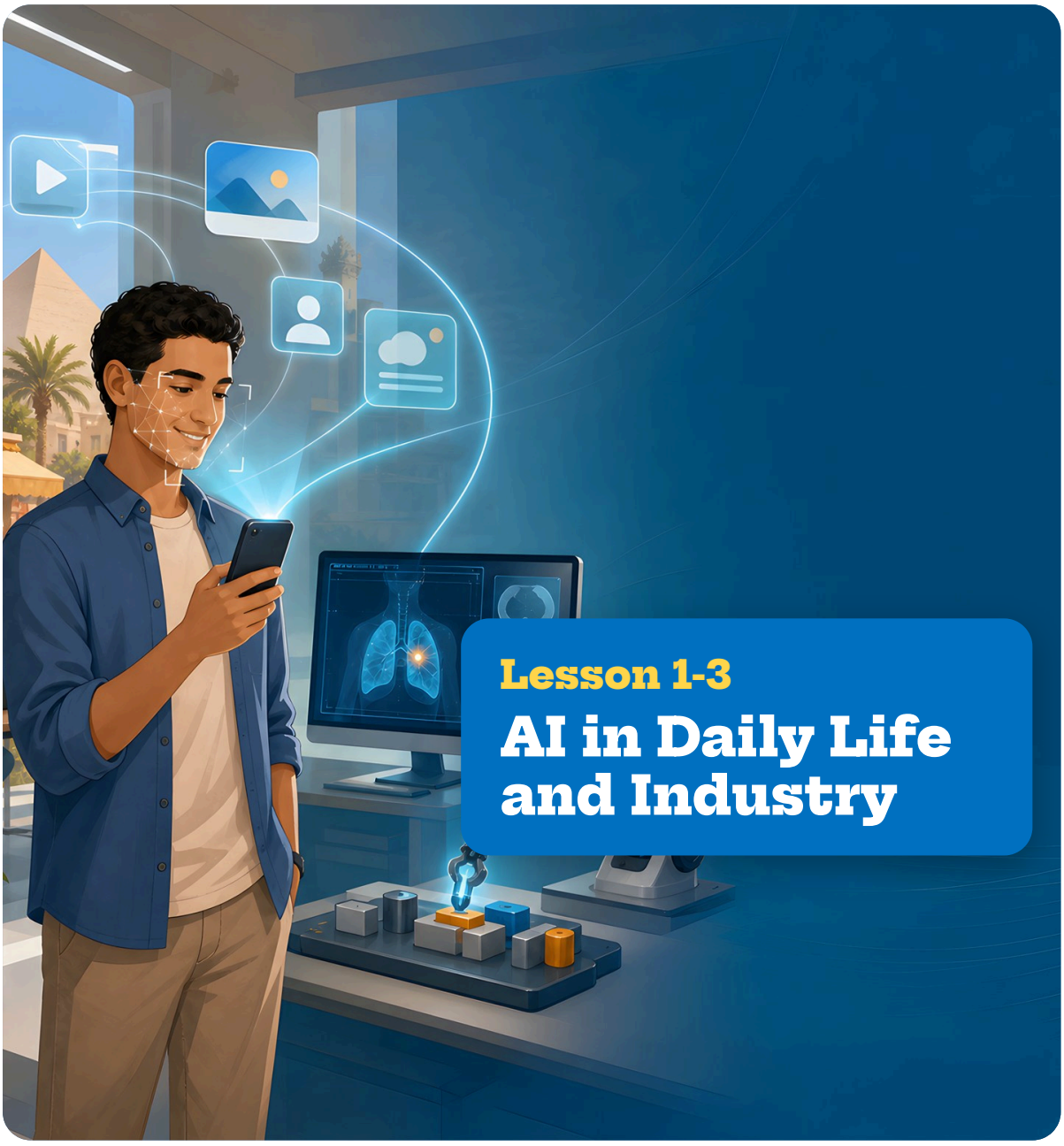
B AI is a type of machine learning, and generative AI is a type of AI.

C Deep learning and machine learning are completely different technologies.

D Generative AI is a technology that generates data without using machine learning.

⭐ Key takeaway

Remember:
AI is the broad field; machine learning sits inside it, deep learning inside machine learning, and today's generative AI is built on deep learning — from the broadest idea to the most specialized.



Learning Objectives

- Give examples of how AI is used in daily life and various industries.
- Explain what AI is good at and what requires caution when using AI.

AI is already part of many everyday services. A video site recommends the next clip, a voice assistant answers a spoken question, a translation tool converts text on a foreign menu, and a phone unlocks by recognizing your face. In industry, AI analyzes medical images, predicts when a machine will fail, and plans delivery routes. AI is effective at identifying patterns in data, but it is not suited to every task, and some uses require caution.

? Guiding question: Where is AI used in daily life and industry, what is AI good at, and what needs caution?

✎ Explore · In Pairs

With a partner, list three services you used this week that you think used AI (for example, a video recommendation, a voice assistant, or a translation). For each, predict what data the AI needed to do its job. Then decide together: which one would matter most if it made a mistake?

Key Fact:
AI is good at finding patterns in data, but human judgment is still needed where ethics, privacy, and responsibility are involved.

Key Concepts:

- recommendation system
- voice assistant
- machine translation
- face recognition
- image-diagnosis AI
- predictive maintenance
- black-box problem
- hallucination

📌 Point!

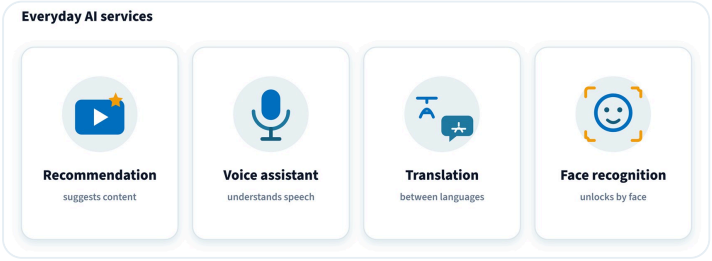
1. Use of AI in Daily Life

(1) Various AI technologies are already in use around us.

Service	Role of AI	Examples
Recommendation systems	Predict preferences from past behavior data and display recommendations.	YouTube, Amazon, Spotify, etc.
Voice assistants	Recognize a voice, understand commands, and execute them.	Siri, Google Assistant, etc.
Machine translation	Automatically translate text into different languages.	Google Translate, DeepL, etc.
Face recognition	Automatically detect and identify human faces in photographs.	Smartphone unlocking, etc.



Recommenders match new items to your past choices.



Four everyday-life AI services

KEY TERMS
recommendation system — a system that predicts preferences from past behavior data and displays recommendations.

2. Use of AI in Industry

(1) AI technologies are also being introduced in various industries.

Industry	Examples of AI use
Healthcare	Image-diagnosis AI (detecting diseases from X-ray and CT images), drug-discovery support
Agriculture	Predicting harvest timing, detecting pests and diseases
Manufacturing	Automation of product quality inspection, predictive maintenance
Logistics	Optimization of delivery routes



Four industries where AI is used

3. Characteristics of AI and Cautions for Use

(1) What AI is good at

- Finding and classifying patterns in complex data such as images and text
- Recognition and generation of images, audio, and text
- Probabilistic reasoning and prediction based on data

(2) What requires caution when using AI

- Ethical judgments (may lead to discrimination or prejudice)
- Judgments involving personal information or privacy
- Final decision-making (who is responsible for the result)
- Accuracy of judgments when training data is biased
- Generating content that differs from the facts (hallucination)
- Issues of rights when using copyrighted works as training data
- Loss of clarity about how a judgment was made (the black-box problem)

✎ Worked Example

(1) For each of a-d, match the AI technology used in the service with the most appropriate term from the choices below (A–D).

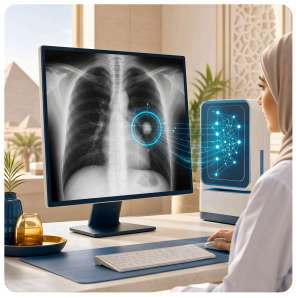
a YouTube recommends videos that match the user's preferences

b Speaking to a smartphone to check the weather

c Translating a foreign-language website into one's native language

d A smartphone camera automatically detects a person's face

[Choices] **A** Recommendation system **B** Voice assistant **C** Machine translation **D** Face recognition



In industry, AI flags what humans might miss.

💡 THINK IT THROUGH

A hospital is considering using image-diagnosis AI to detect diseases from X-ray images. Explain why the final diagnosis should be confirmed by a human doctor rather than left to AI alone.

💡 PAUSE & THINK

AI services often use your personal data — what you watch, buy, or say. Why does this raise privacy concerns, and who should decide how your data is used?

KEY TERMS

black-box problem — when it is unclear how an AI reached its judgment.

(2) Mark each statement (A–D) with ○ if true or × if false.

A AI is good at probabilistic reasoning and prediction based on data.

B AI is good at recognizing images and audio.

C It is fine to leave ethical judgments entirely to AI.

D AI judgments can become inaccurate when training data is biased.

✓ Solution

(1)

a: Recommendations matched to preferences come from a recommendation system. **A**

b: Operating by speaking is a voice assistant. **B**

c: Translating a foreign language into one's native language is machine translation. **C**

d: Detecting a person's face is face recognition. **D**

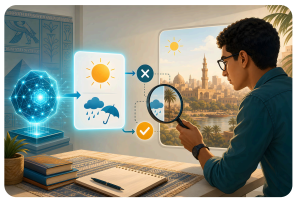
(2)

A AI is good at probabilistic reasoning and prediction based on data. Therefore, ○.

B AI is good at recognizing images, audio, and text. Therefore, ○.

C AI judgments can lead to discrimination or prejudice, so ethical judgments should not be left entirely to AI. Therefore, ×.

D Bias in training data affects the accuracy of AI judgments. Therefore, ○.



AI can be confidently wrong — verify its output.

Try

1 Answer the following questions.

1. What is the term for the system that predicts preferences from past behavior data and displays recommendations?
2. What is the term for the AI technology that recognizes a voice, understands commands, and executes them?
3. In which industry are image-diagnosis AIs used?
4. In manufacturing, what is the term for the system that predicts product failures in advance?

EXAM-STYLE QUESTION

Analyze why AI is well suited to some tasks but requires caution in others. **[6] In your answer:** refer to what AI is good at (finding patterns, recognition, prediction) and to what needs caution (ethical judgments, privacy, final decision-making).

2 Answer the following questions.

(1) From the following options (A–D), choose the one that does NOT belong to "what requires caution when using AI."

A Ethical judgments

B Finding and classifying patterns

C Judgments involving personal information or privacy

D Final decision-making

(2) For each AI usage example (a–d), indicate which industry is most closely related: **A** Healthcare, **B** Agriculture, **C** Manufacturing, **D** Logistics.

a Optimizing delivery routes

b Detecting diseases from X-ray images

c Detecting pests and diseases

d Automating product quality inspection

⚙ Think as an Engineer — Investigate & Decide

You have just discussed agriculture in the Think It Through note. Now investigate a different industry and decide.

1 · Investigate. Choose one industry from the lesson other than agriculture — healthcare, manufacturing, or logistics. Find one real, documented example of AI being used in it, and record the source you used.

2 · Weigh it up. Note one benefit and one caution of using AI in that industry, using the ideas of what AI is good at and what needs caution.

3 · Decide. Should this industry use AI for this task on its own, or with a human checking the result? Recommend one approach and give two reasons.

Stuck? AI is good at finding patterns quickly; but a human is needed where a wrong result could harm someone or involve responsibility.

🌍 In a New Context

A school library wants to use AI to recommend books to students based on what they have borrowed before. Using what you learned about recommendation systems and what AI is good at, explain how the AI would work, and identify one caution the school should consider.

💡 THINK IT THROUGH

In agriculture, give one realistic advantage of introducing AI and one concern that should be considered before introduction, and briefly explain the reason for each.

❓ Lesson Question — Answered

AI is used widely in daily life — in recommendation systems, voice assistants, machine translation, and face recognition — and in industry, across fields such as healthcare, agriculture, manufacturing, and logistics. AI is good at finding and classifying patterns in data, recognizing and generating images, audio, and text, and making probabilistic predictions from data. However, it needs caution where ethical judgments, personal information and privacy, and final decision-making are involved, and where training data is biased. So AI is a powerful tool for pattern-based tasks, but human judgment is still needed for responsibility, ethics, and fairness.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Answer the following questions.

(1) Select all options that AI excels at.

A Ethical judgments

B Finding features in images and text and classifying them

C Judgments involving personal information

D Probabilistic reasoning and prediction based on data

E Decision-making that bears responsibility for results

F Recognition and generation of images, audio, and text

(2) From the following options (A–D), choose the one that is NOT an appropriate description of services that use AI.

A A recommendation system predicts preferences from past behavior data and displays recommendations.

B A voice assistant recognizes a voice, understands commands, and executes them.

C Machine translation is a system in which humans manually translate foreign languages.

D Face recognition automatically detects and identifies human faces in photographs.

3 Read the following passage and answer each question.

AI is widely used in our daily lives. For example, recommendations of videos that match one's preferences on a video site come from a (a) system. Also, AI

⭐ REFLECT & CHALLENGE

Reflect: Which use of AI in this lesson surprised you most? Where in your own life do you think AI is making decisions about you? **Challenge:** Choose one task where AI needs caution (ethics, privacy, or final decisions). Suggest one rule that would make using AI for that task safer.

technology is used in (b), with which one can speak to a smartphone to operate it, and in (c), which converts foreign-language text into one's native language.

1. Fill in the blanks (a)–(c).

(2) From the following (a–d), choose all that require caution when using AI.

a High-speed processing of large amounts of data

b Ethical judgments

c Judgments involving personal information or privacy

d Recognition of images and audio

⭐ Key takeaway

Remember:
AI is good at finding and classifying patterns in data — recommendations, voice, translation, face recognition, and many industry tasks — but human judgment is still needed where ethics, privacy, and responsibility are involved.



Lesson 1-4

Ethical Issues with AI

Learning Objectives

- Explain what algorithmic bias is and describe its main causes.
- Connect the basic principles of AI ethics (fairness, transparency, privacy protection, and accountability) to specific situations.

AI now helps make real decisions about people: selecting candidates for job interviews, identifying individuals through face recognition, and determining how personal data is used. But an AI can be unfair if the data it learned from was biased, and it can be hard to know why it made a decision at all. As AI spreads, questions arise about how to keep it fair, transparent, and accountable, and about who is responsible when it produces an incorrect or unfair result.

? Guiding question: What ethical issues arise as AI spreads, and what principles should guide how we use it?

✎ Explore · In Pairs

With a partner, look at this situation: a face-recognition AI is more likely to misidentify people from some ethnic groups than others. Before you read on, predict who could be harmed by this, and how. Give a reason.

Point!

1. Algorithmic Bias

(1) **(Algorithmic bias)**…bias in AI judgments caused by bias in the training data. *e.g., a hiring AI unfairly evaluating a particular gender; a face-recognition AI being more likely to misidentify particular ethnic groups*

(2) Main causes of bias

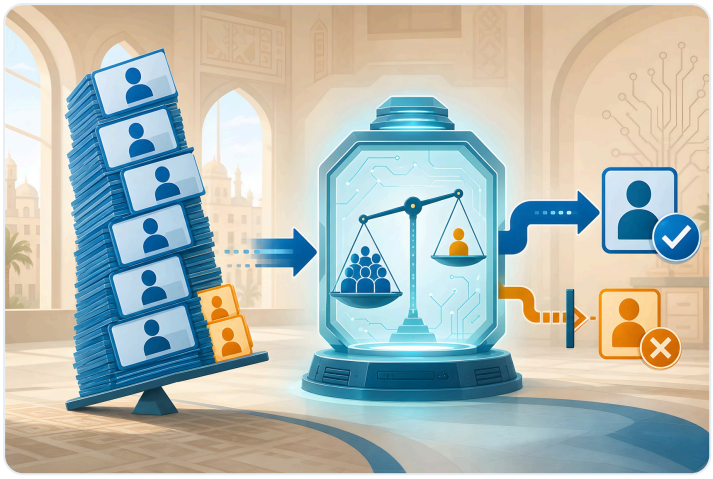
- Training data is biased (e.g., insufficient data on certain attributes; past discriminatory tendencies are reflected in the data). Bias can also enter through the way the system is built — for example, an inappropriate choice of variables, a proxy variable that indirectly stands in for a protected attribute, or the design of the model itself.

Key Fact:
If the training data is biased, an AI can reproduce that bias in its judgments; using AI responsibly means checking for bias, being able to explain decisions, and knowing who is accountable.

Key Concepts:

- algorithmic bias
- training data
- face recognition
- privacy
- explainable AI (XAI)
- black-box problem
- responsibility
- fairness
- transparency
- privacy protection
- accountability

KEY TERMS
algorithmic bias — bias in AI judgments caused by bias in the training data.



Biased training data leads to biased AI judgments

2. Privacy Issues

(1) The development of AI technology has given rise to new issues concerning **(privacy)**.

① Surveillance through face-recognition technology…cameras in public spaces can identify and track individuals.

② Mass collection of personal data…large amounts of online behavior data are collected and analyzed.

3. Explainable AI (XAI) and Responsibility

(1) **(Explainable AI (XAI))**…technology that makes it possible for humans to understand why an AI made a particular judgment.
When an AI's decision-making process is opaque (a black box), it is difficult to verify whether the result is correct.

(2) **(Responsibility)**…the question of who is responsible when an AI makes an incorrect judgment.
Views on responsibility differ depending on whether one is the developer, the user, the operator, etc.

4. Basic Principles of AI Ethics

(1) The following basic principles are considered important for using AI appropriately.

Principle	Description
Fairness	Not unjustly discriminating against any particular person or group
Transparency	Showing the AI's decision-making process and inner workings clearly
Privacy protection	Handling personal information appropriately and protecting privacy
Accountability	Being answerable for the AI's decisions

THINK IT THROUGH

When face-recognition technology is used in public spaces, safety and convenience may improve, but privacy concerns also arise. Explain how convenience, safety, and privacy protection should be balanced, and state which basic principle of AI ethics is most closely related to your view.



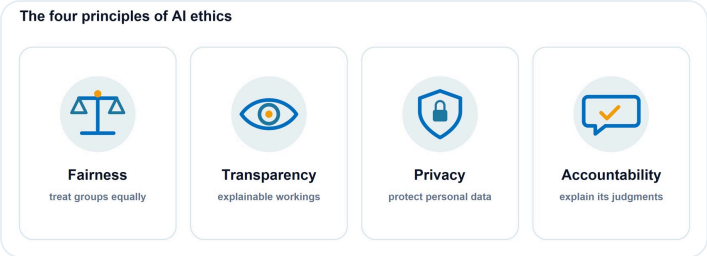
Privacy: control which personal data is shared.

KEY TERMS

explainable AI (XAI) — technology that makes it possible for humans to understand why an AI made a particular judgment.



Explainable AI reveals the reasons behind a decision.



The four principles of AI ethics

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Algorithmic bias is bias in AI judgments caused by bias in the training data.

B Even if an AI's decision-making process is opaque, there is no problem as long as the result is correct.

C Explainable AI (XAI) is a technology that makes it possible for humans to understand the AI's reasoning.

D When an AI makes an incorrect judgment, who is responsible has already been clearly determined.

(2) Match each description (a–d) with the most appropriate AI ethics basic principle from the choices below (A–D).

a Not unjustly discriminating against any particular person or group

b Showing the AI's decision-making process and inner workings clearly

c Handling personal information appropriately and protecting privacy

d Being answerable for the AI's decisions

[Choices] **A** Fairness **B** Transparency **C** Privacy protection **D** Accountability

✓ Solution

(1)

A Correct definition of algorithmic bias. Therefore, ○.

B When an AI's decision-making process is opaque, it is difficult to verify whether the result is correct. Therefore, ×.

C Correct description of Explainable AI (XAI). Therefore, ○.

D Views on responsibility when an AI makes an incorrect judgment differ depending on whether one is the developer, the user, the operator, etc., and have not yet been clearly determined. Therefore, ×.

(2)

a: Not discriminating is fairness. **A**

PAUSE & THINK

If no one can explain why an AI rejected someone's job application, is that fair? Which principle of AI ethics is missing?

KEY TERMS

fairness — no unjust discrimination against a person or group.
accountability — being answerable for the AI's decisions.

b: Showing the decision-making process is transparency. **B**

c: Protecting privacy is privacy protection. **C**

d: Being answerable for decisions is accountability. **D**

Try

1 Answer the following questions.

1. What is the term for the phenomenon in which bias arises in AI judgments due to bias in the training data?
2. What is the term for the technology that makes it possible for humans to understand why an AI made a particular judgment? Also, give its three-letter abbreviation.
3. Among the basic principles of AI ethics, what is the term for the principle that requires not unjustly discriminating against any particular person or group?
4. Among the basic principles of AI ethics, what is the term for the principle that requires handling personal information appropriately?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate cause of algorithmic bias.

A Training data is biased.

B Past discriminatory tendencies are reflected in the data.

C The processing speed of the computer is slow.

D There is insufficient data on certain attributes.

(2) From the following options (A–D), choose all that are privacy issues associated with AI.

A Individuals are identified and tracked through face-recognition technology.

B Online behavior data is collected and analyzed in large quantities.

C AI creates creative works.

D AI computation speed improves.

Think as an Engineer — Investigate & Decide

Building on the hiring-AI case in the Think It Through note, investigate a real example and decide on a rule.

1 · Investigate. Find one documented case where an AI system was accused of being unfair or biased, and record the source you used (a news report, study, or official statement). What group did it affect, and what data might have caused it?

KEY TERMS

privacy — appropriate handling and protection of personal data.
responsibility — who answers when an AI judgment is wrong.

EXAM-STYLE QUESTION

Analyze how algorithmic bias can arise in an AI system, and how it can be reduced. **[6] In your answer:** refer to the cause of bias (biased or insufficient training data) and to how the principles of fairness and transparency help address it.

2 · Consider responsibility. When an AI makes an unfair decision, who is responsible — the developer, the company using it, or the operator? Note the responsibility each one holds.

3 · Decide. Before a company uses a hiring AI, what one rule should it follow to make the system fairer? Justify your rule using the principles of AI ethics.

Stuck? If the training data reflects past bias, the AI can repeat it — one fix is to check the data and test the AI's results across different groups.

🌍 In a New Context

A school wants to use face-recognition cameras at its gate to record attendance automatically. Identify one practical benefit for the school and one privacy concern. Then, if the camera misidentifies a student, who should be responsible — the school, or the company that made the AI? Give a reason.

❓ Lesson Question — Answered

The lesson question asked what ethical issues arise as AI spreads and what principles should guide how we use it. Because AI now makes decisions that affect people, three concerns stand out: an AI can be unfair when it learns from biased data; it can put people's privacy at risk; and it can be difficult to understand, or to hold anyone answerable for, its decisions. Handling these well takes more than accuracy. The lesson sets out four principles to aim for — fairness, transparency, privacy protection, and accountability — and keeping AI fair, transparent, privacy-respecting, and accountable is what makes its use responsible.

💡 THINK IT THROUGH

A hiring AI was found to rate applicants of a particular gender lower. Using the ideas of algorithmic bias and the basic principles of AI ethics, explain the most likely cause and one possible measure to address it. There can be more than one reasonable view. Give the basis for your answer.

KEY TERMS

transparency — showing the AI's decision-making process clearly.
privacy protection — handling personal information appropriately.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

As AI technology spreads widely in society, attention has been drawn to various ethical issues. For example, there is the issue of (a), in which bias arises in AI judgments due to bias in the training data. Also, for the "black-box" problem—where it is unclear why an AI made a particular judgment—a technology called (

b) is being researched. Furthermore, when an AI makes an incorrect judgment, (c) is questioned, but no clear standard has yet been established among developers, users, and operators.

1. Fill in the blanks (a)–(c).

3 Answer the following questions.

(1) For each of a-d, indicate which AI ethics basic principle is most closely related: **A** Fairness, **B** Transparency, **C** Privacy protection, **D** Accountability.

a Ensuring that a hiring AI evaluates fairly regardless of gender.

b Disclosing the AI's decision-making process to users in an easy-to-understand way.

c Not using collected personal data for purposes other than as originally intended.

d Being answerable when an AI's diagnostic result turns out to be wrong.

(2) From the following options (A–D), choose the one that most appropriately describes ethical issues with AI.

A There are no ethical issues with AI.

B Algorithmic bias is caused by the AI's processing speed.

C Bias in training data can cause discriminatory bias in AI judgments.

D Even if an AI's decision-making process is opaque, there is no problem for users.

⭐ REFLECT & CHALLENGE

Reflect: Has this lesson changed how much you would trust an AI decision about you? Which of the four principles do you think is hardest to guarantee, and why?
Challenge: Choose one place AI is used (hiring, face recognition, or recommendations). Suggest one rule based on the AI ethics principles that would make it more trustworthy.

⭐ Key takeaway

Remember:
If training data is biased, an AI can reproduce that bias; using AI responsibly means checking for bias, being able to explain decisions, and knowing who is accountable — guided by fairness, transparency, privacy protection, and accountability.